

Agentic AI in Social Science Research

From Annotators to Autonomous Pipelines

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Where I Am Coming From

The day job

- Political economist, and a generalist by temperament.
- Associate editor, *International Studies Review*.
- Founder of the **Politics of AI** MSc at Leiden.
- Founded an education AI start-up.

Two ways into AI

- **Substantively**: what it does to politics and to policy.
- **Methodologically**: what it does to how we do research.

Also, honestly

I use these tools all day outside the university as well: running a business, and raising children.

That matters for the talk. Most of what I know about where agents break, I learned by relying on them for something I could not afford to get wrong.

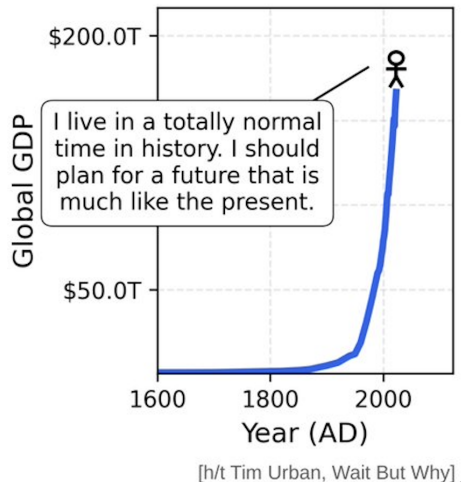
Nothing here is speculative. It is what I actually do, including the parts that went badly.

From LLM-as-Tool to LLM-as-Colleague

A year ago: LLMs *annotate, classify, scale* text.

Today: LLMs *plan, search, call code, call other LLMs, and deliver finished pipelines.*

- The unit of work is no longer “one prompt, one answer.”
- It is a multi-step research workflow with checkpoints.
- Researchers act as **managers**, not typists.



Roadmap

- 1 What the Machine Can Do, and What Is Left for Us
- 2 What Is an Agent?
- 3 Case Study: Economic Exposure and Congressional Rhetoric
- 4 Is This Legitimate?
- 5 What the Pipeline Taught Me
- 6 Other Uses of Agents
- 7 From Adaptive Surveys to Agentic Interviewing
- 8 From Synthetic Individuals to Simulated Societies
- 9 The Research Lifecycle With an Agent in the Loop
- 10 Building Your Own Pipeline
- 11 Ethics
- 12 Appendix

I am also happy to answer questions on my set up.

The Paper Machine Exists

Three data points from the last 18 months.

- **Sakana's AI Scientist-v2**: an AI-written paper passed double-blind review at an ICLR 2025 workshop.
- **Agents4Science** (Stanford, Oct 2025): AI authors, AI reviewers. 315 submitted, 48 accepted, 5 fully AI-generated.
- **Kohler et al. (2026)**: agents reproduce published results from the methods section and the data, never the code.

A paper-shaped object is now cheap.

Do we hand over the process? No. Hand over the **labour**, keep the **judgment**, inspect **every output**.

The AI Scientist Generates its First Peer-Reviewed Scientific Publication

March 12, 2025

[CS](#) [AI](#) [ML](#) [DL](#) [NLP](#) [CV](#)

Open Conference of AI Agents for Science 2025

The 1st open conference where AI serves as both primary authors and reviewers of research papers

Computer Science > Artificial Intelligence

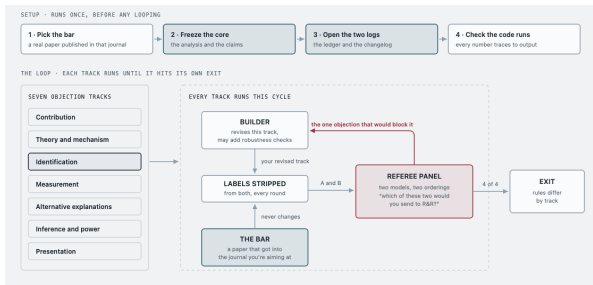
(Submitted on 23 Apr 2026)

Read the Paper, Write the Code: Agentic Reproduction of Social-Science Results

Benjamin Kohler, David Zollkofer, Johanna Einsiedler, Alexander Hoyle, Elliott Ash

Recent work has used LLM agents to reproduce empirical social science results with access to both the data and code. We broaden this scope by asking: Can they reproduce results given only a paper's methods description and original data? We develop an agentic reproduction system that extracts structured methods descriptions from papers, runs reimplementations under strict information isolation — agents never see the original code, results, or paper — and enables deterministic, cell-level comparison of reproduced outputs to the original results. An error attribution step traces discrepancies through the system chain to identify root causes. Evaluating four agent scaffolds and four LLMs on 48 papers with human-verified reproducibility, we find that agents can largely recover published results, but performance varies substantially between models, scaffolds, and papers. Root cause analysis reveals that failures stem both from agent errors and from underspecification in the papers themselves.

My Counter-Move: The Social Science Gauntlet



A real published paper as the bar, identifying detail stripped from both. Seven objections, not sections. Two referees from **different model families**, each comparison run in both orders: a win is four out of four. Headline frozen, every check logged. Back come the revised draft, every change that touched a claim, and the objections it *could not* answer.

github.com/matthewdiguseppe/social-science-gauntlet-loop

The machine attacks the paper. It cannot touch the result.

My Impressions, a Year In

Good at

- Running the robustness checks. All of them, overnight.
- Acting as a referee: finding the objection you did not want to hear.
- The labour: scraping, cleaning, coding, logging.

Not yet

- Writing. The prose is fluent and forgettable.
- Judging which tests belong in the paper, and which do not.
- Knowing what the field will find interesting.

So you still have to be the expert.

The machine checks the paper. It cannot tell you what the paper should be. Somebody in the loop has to know what they are doing, and for the students in the room that somebody is you.

You cannot offload the judgment wholesale. Take the time to learn the craft. The agent makes that craft worth more, not less.

The Comparative-Advantage Map

Stage	Agent takes the labour	Human keeps the judgment	The gate: what you inspect
Ideate	Literature scan, alternative angles, scoop check	The question and why it matters	The shortlist, not the summary
Design	Draft codebook, simulate respondents, power calculation	Construct, identification, ethics	Hand-code 30–50 cases yourself
Build	Scrape, clean, code, log, checkpoint	The spec and the stopping rule	What changed, and the failure register
Analyse	Run the specs, robustness, figures	Model choice, interpretation	The ledger, not the agent's summary
Write	Style, structure, consistency checks	The argument, the claims, the numbers	Every sentence that contains a number
Submit	Journal fit, mock referee, guideline check	Which journal, what to concede	The referee report as a stress test

This talk is about the gate column.

If a stage produces nothing you can inspect, you have delegated judgment, not labour.

What Can Agentic AI Do for Your Research?

- Build datasets from the open web or existing datasets/corpora at scale
- Run multi-stage pipelines unattended
- Code, debug, and document its own scripts
- Conduct adaptive interviews and surveys
- Simulate respondents to pre-test designs
- Populate small-scale societies for ABM-style work
- Produce a full audit trail by default

The frontier is not what an agent can do — it is what we can *trust* an agent to do.

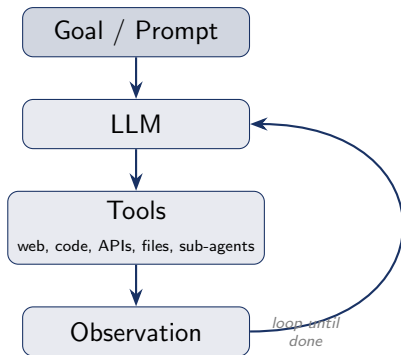
The Minimal Definition

An **agent** is an LLM that:

- 1 Receives a goal
- 2 Selects and invokes **tools** (web search, code execution, file I/O, APIs, other LLMs)
- 3 Observes the result
- 4 Iterates until done (or hits a stop condition)

What changes vs. a chatbot?

- Persistent state across many turns
- Tool use $\Rightarrow$ the model can *act*, not just *speak*
- Self-correction inside one task



The Harness: Everything Around the Model

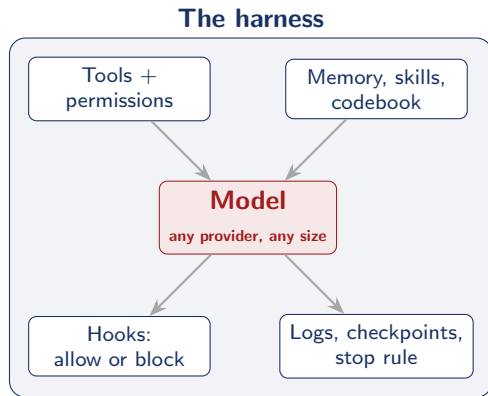
The model is the engine. The harness is the car.

The harness runs the loop: holds the conversation, hands the model its tools, checks what it may do, logs every step, decides when to stop.

What lives in the harness

- Tools, and the permissions on each.
- Standing context: memory, skills, codebook.
- Hooks that can block an action.
- Logs, checkpoints, the stop rule.

Claude Code, Codex CLI, pi, OpenCode: harnesses. So is a custom GPT, with an uploaded corpus as its context: the top half of this diagram. This talk is the bottom half.



Your methodology lives in the harness. The model is replaceable; the audit trail is not.

The Loop, One Turn at a Time

One turn

- 1 The model reads the context: the goal plus everything so far.
- 2 It either answers or asks for a tool call.
- 3 The harness runs the tool and appends the result.
- 4 Back to step 1.

Until a stop condition

- The model says it is done.
- A cap on turns, tokens, or money.
- A hook says stop.

A real trace: Hormuz, Tier 0, one legislator

```
t1 model -> web_search("Doe" Iran Hormuz 2026)
   tool  <- 8 URLs with titles and snippets
t2 model -> web_search("Doe" Iran statement 2026)
   tool  <- 6 URLs
t3 model -> JSON {queries_issued, urls_found}
   harness: stop. Done, and 2 queries = the cap.
```

```
pipeline: fetch + archive the 14 URLs
          extract (no internet) -> verify
          0 verified quotes? new call, Tier 1
```

Two loops. The inner loop is the model's: which tool to call next and what to do with what comes back. In Hormuz even the queries are prescribed. The outer loop is yours: over 538 legislators, and up the tier ladder when a call returns nothing.

Every decision inside the loop is the model's. Every boundary around it is the harness's.

Hooks: Code That Runs Inside the Loop

A prompt asks. A hook enforces.

A hook is a script the harness runs at a fixed point in the loop: before a tool call, after it, or when the agent wants to stop. It sees the call and can allow it, block it, or edit it. It can also write to a log.

Hooks worth having

- **Cost guard.** Before any batch submission: no approved cost audit, no run.
- **Ledger.** After every tool call: append what ran and what came back.
- **Scope.** Block writes outside the project folder. Block deletes.
- **Checks.** After every edit, run the tests. After coding, run the verifier.
- **Stop.** After N calls or $\$X$, stop and ask.

The one on my laptop

Registered in settings.json:

```
"PreToolUse": [{  
  "matcher": "Bash",  
  "hooks": [{ "type": "command",  
    "command": "python3 cost_guard.py" }]  
}]
```

Before every shell command, the script reads the call. If the call looks like an LLM batch submission and no approved cost audit sits next to the input file, the hook exits with code 2. The command never runs, and the agent is told to run the audit first.

The model cannot talk its way past a hook. That is the whole point.

The Question

The event: On 28 February 2026, the US launched military operations against Iran. The Strait of Hormuz effectively closed shortly after.

The puzzle: Economic costs were **not distributed evenly** across US Congressional districts.

- Petrochemicals, freight logistics $\Rightarrow$ acute disruption
- Finance, education, legal services $\Rightarrow$ insulated

The hypothesis: Legislators from more exposed districts express **greater opposition** to the conflict.

Core prediction

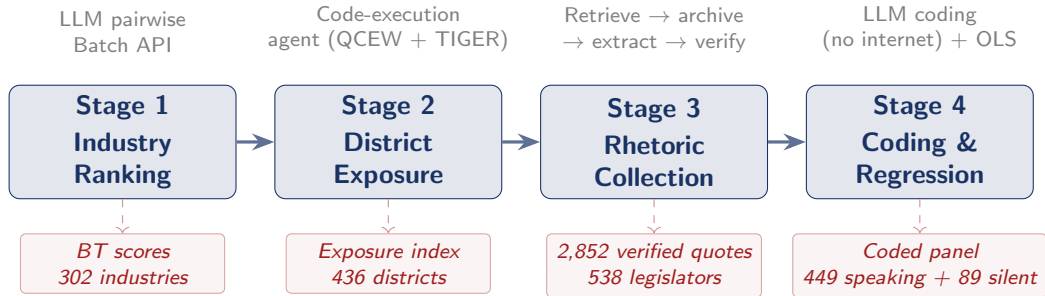
$$\underbrace{\text{Exposure}_d}_{\text{district}} \Rightarrow \underbrace{\text{conflict support}_i}_{\text{legislator}}$$

$$\beta_{\text{exposure}} < 0$$

The challenge

Build, code, and analyse rhetoric for the entire House and Senate — within *days* of the event, by myself.

A Four-Stage Agentic Pipeline



Agents at every stage. The researcher writes the spec; agents do the labour.

Each stage is a checkpoint. Pipeline is resumable; failures are local.

Zoom In on Stage 3 — Where the Agents Live

The seduction. A single API call with a built-in web tool returns a coded political position in 30 seconds for a few cents.

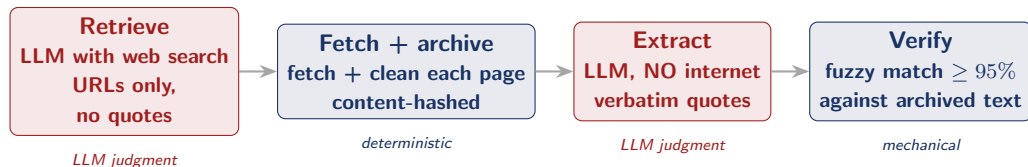
What previously needed a trained RA team, an inter-rater protocol, and weeks of coding can now run *overnight on a laptop*.

The danger

That same single call:

- **Logs no queries** — you can't replicate what was searched.
- **Returns no source text** — nothing to verify the quote against.
- **Cannot distinguish** “no statement exists” from “the model invented one.”
- **Conflates retrieval and extraction** — one black box does both.

The Architecture: Retrieve, Archive, Extract, Verify



The rule: the LLM is allowed to *search*, but not to pretend it has *read* what it searched.

- Retrieval call returns queries and URLs only — no quotation, no synthesis.
- Every URL fetched independently, archived as raw HTML + cleaned text, each with a content hash: a fingerprint that changes if the text does.
- Extraction call has no internet — it can only read the archive.
- Every extracted quote mechanically checked against the archive; failures dropped *before* coding.

The Stopping Rule: When the Agent Stops Searching

Each tier is a separate agent call with its own prompt.

The agent issues only the queries the current prompt tells it to. The *pipeline* decides whether to issue the next-tier prompt.

Three exit conditions:

- 1 **Success.** ≥ 1 verified quote in the current tier $\rightarrow$ stop, code the quote(s).
- 2 **Failure.** Tier T4 returns 0 verified quotes $\rightarrow$ emit *no recoverable statement*.
- 3 **Resource cap. 17 queries per legislator** across all tiers (2+3+4+4+4).

The agent never says “one more thing.” The pipeline owns the stop. The stop decides who is silent.

The escalation ladder:

Tier	Q	Triggered when	Search strategy
T0	2	always	Canonical: <i>name + Iran Hormuz 2026</i>
T1	3	$T0 = 0$	Name variants
T2	4	$T1 = 0$	Channel: <i>site:.gov</i> , socials
T3	4	$T2 = 0$	Proxy events: <i>votes</i> , letters
T4	4	$T3 = 0$	District press
T5	0	$T4 = 0$	Stop — emit silence

Q = number of web searches in that tier's prompt.

Inside the Extract Call

You are extracting verbatim quotes from a single archived news article attributed to a specific US legislator.

You DO NOT have internet access. The only text available is what is included below under SOURCE TEXT. If a fact, date, or quote is not in that text, you do NOT know it. Do not invent.

SOURCE TEXT

```
=== BEGIN SOURCE TEXT ===  
{source_text}  
=== END SOURCE TEXT ===
```

TASK

Identify EVERY passage that is a direct quote attributed to {full_name} OR a clear paraphrase of a statement they made about the Iran / Hormuz crisis. For each, output:

- claimed_quote: EXACT verbatim chars from source
- offset_start / _end: 0-indexed character offsets
- is_direct_quote: bool
- claimed_date: YYYY-MM-DD or null (do NOT guess)
- context: 1-sentence note (<=200 chars)

ABSOLUTE RULES

- claimed_quote MUST appear verbatim; downstream verification rejects any quote at fuzzy ratio < 95.
- Do NOT paraphrase or normalise punctuation.
- Do NOT use prior knowledge of the legislator or event.
- Output JSON only.

The Codebook: What the Agents Actually Apply

The research instrument.

Each verified quote scored 1–5 on *conflict support*:

- 1 Strongly opposes
- 2 Somewhat opposes
- 3 Neutral / procedural
- 4 Somewhat supports
- 5 Strongly supports

Plus 6 binary frames: economic cost, security, constituent impact, admin praise, admin criticism, diplomacy.

Building the codebook **is** the research design.

- ① Draft v0 from the construct in the literature.
- ② Hand-code 30–50 random cases; revise rules where your intuition disagreed. Tuning, not validation.
- ③ Add worked examples for the hard cases; write an explicit “*if ambiguous, do X*” rule.
- ④ Pilot on the LLM; compare to your hand-codes; revise.
- ⑤ Lock the prompt, hash it, pin the model.

The Validity Checks

Mechanical verification

Every extracted quote must match the archived page text at $\geq 95\%$ **character similarity**.

2,910 extraction attempts $\rightarrow$ **2,852** verified $\rightarrow$ **58** rejected = **2.0%** failed verification.

Reviewers don't have to trust the LLM — the verifier doesn't either.

Coverage as a missing-data mechanism

449 speaking + **89** silent.

Each silent member carries a coverage reason, so the mechanism can be modelled. The headline estimand is conditional on speaking.

Failure register (no free-text notes)

Page fetch failed	2,198
Source text empty	490
Bot challenge	303
Paywall / forbidden	146
Quote not found in the archived text	58
Flagged for human review (no rule fired)	0

Identifier lineage

Every artefact is content-hashed.

query $\rightarrow$ URL $\rightarrow$ raw page $\rightarrow$ cleaned text $\rightarrow$
extraction $\rightarrow$ verified quote $\rightarrow$ coding

A reviewer can re-derive the chain.

The Numbers

Scale

- **538** legislators (House + Senate)
- **2,852** verified quotes (avg 6.4 per speaking member)
- **6,899** LLM API calls in the rhetoric stages
- **1.5 hours** wall-clock, 8 parallel workers

Cost

- Total: **\$62.58**
- Comparable RA-coded study: \$1,500
- Pipeline is $\sim$ **24** $\times$ **cheaper**

Quality

2.0% of extracted quotes failed verification.
Zero items flagged for human review.
Every quote traceable to its archived source.

Substantive result (illustrative)

Pooled OLS, controls: $\beta_{\text{exposure}} = +0.035$,
 $p = 0.258$ — **null**.

Most robust within-party finding: asymmetric *constituent-impact framing* — exposed Dems invoke it more, exposed Reps invoke it less.

The methods talk doesn't depend on which way the coefficient points.

DV: member mean of quote-level codes (1–5),
conditional on speaking.

What's the Right Counterfactual?

The methods literature usually picks the wrong comparison.

Wrong counterfactual #1: deterministic text-as-data

Dictionary methods, supervised classifiers, BERT — reproducible, auditable, transparent.

Against this benchmark the LLM looks like a black box.

Most coding tasks don't benefit from these tools.

Wrong counterfactual #2: an idealised RA team

Trained coders, validated protocol, second coder, κ reported.

Against this benchmark the LLM looks underregulated.

But this is not what most projects actually do.

And in 2026: can you swear your RAs aren't quietly running everything through ChatGPT themselves?
Hidden cyborgs everywhere. The idealised RA team is a relic of a pre-2023 world.

The Honest Counterfactual: One RA, Also a Black Box

The actual alternative is one RA reading and coding texts.

That is a **black box too**.

- Coding rules drift across the day.
- Fatigue effects are unmeasured.
- No second coder, no κ reported.
- Decisions are not logged.
- Replication is impossible — the RA graduated.

We trusted that black box for decades.

The honest question is not
“Is the LLM a black box?”

It is
“*Which* black box,
and which one can we audit?”

The Access Argument

Even the RA comparison overstates who has the resource.

Who has no RAs?

- Teaching-focused institutions
- Researchers without grant funding
- Early-career scholars
- Most of the Global South
- Independent and policy researchers

Their counterfactual is **a smaller N , a narrower question, or no study at all.**

Dimension	Human RA	Agent
Cost	\$15–25/hr × hrs	\$8 to extract and code; \$63 all-in (Hor-muz)
Availability	Grant-dependent	Anyone with API key
Consistency	Drift, fatigue	Same prompt, same rules
Documentation Scale	Rarely logged 50–200 cases	Full audit trail 538 in 90 minutes
Errors	Random	Systematic bias
Validation	Rarely	<i>Should be</i>

The Honest Comparison

The validation we apply to LLMs —

- existence checks (refetch the URL, verify the quote)
- prompt sensitivity (does the answer survive a rewording?)
- inter-model κ (does Claude agree with GPT?)

— is **stricter** than what is standard for human-coded data.

That is a feature of the method,
not a concession about its limitations.

Lesson 1: Pilot One Agent Before You Launch 500

What happened.

- 1 Launched 10 parallel sub-agents to run the legislator searches.
- 2 All 10 returned **empty results**.
- 3 Cause: **sub-agents do not inherit the parent's tool permissions**. The web-search tool was silently disabled.
- 4 ~20 minutes burned before I noticed.

The fix. Move the search into the parent process; one explicit tool call per legislator; rate-limit; checkpoint to disk.

Pre-flight checklist before any swarm

- Run one agent end-to-end with *explicit* tool calls.
- Verify tool permissions propagate to sub-agents.
- Fix the output format; pilot on 3 cases.
- Only then scale.

Checkpoint and resume

Persist raw results every 25 records. On restart, skip done IDs.

Restartable at zero cost.

Lesson 2: Treat Your Agent Like a Hungover RA

A bright but unreliable RA shows up Monday morning. They will work fast, follow instructions literally, and confidently invent things they don't know.

You wouldn't publish from their first pass without:

- Spot-checking the citations
- Rephrasing the brief and seeing if the answers move
- Having a second RA recode a subsample

Apply the same standard to your agent. The methodology literature treats this as best practice for LLMs — it is just the standard we *should* have been applying to RAs all along.

If your finding doesn't survive a different prompt and a different model, it isn't a finding — it's a prompt artefact.

The three checks — non-negotiable

- 1 **Existence check** (hallucination audit). Refetch every cited URL; verify the quote appears. **<5% unverifiable.**
- 2 **Prompt stability.** Re-run with 2–3 query variants. Report cross-variant DV correlation. **>70% statement overlap.**
- 3 **Inter-model reliability.** Recode with a second model family (Claude/Gemini if you used GPT). **Cohen's $\kappa > 0.70$.**

Reliability Is Not Validity

Reviewer question one: agrees with whom?

Lesson 2's three checks are **reliability** checks. They ask whether the pipeline agrees with itself.

- **Existence:** quote verified, score not.
- **Stability:** a mistake can be stable too.
- **Inter-model κ :** two models agreeing is a fact about the models, not about the world.

Model families share corpora and defaults, so a shared blind spot looks exactly like agreement.

Pangakis, Wolken & Fasching (2023): performance varies by task. Validate every task.

The fix: a held-out human-coded set

- 1 Draw a few hundred cases at random, stratified by party and by search-ladder tier.
- 2 Code them yourself, blind to the model. Never use them to revise a prompt; the 30–50 codebook cases are training data now.
- 3 Report agreement *per category*: κ or F1 for each class, not one overall accuracy.

The recall audit, still to do

The pipeline verified what it found, never what it did *not* find. Hand-search a random sample of the **89 silent**. The share found speaking is the false-silence rate. Report it beside the 2.0% verification failure rate.

Reliability says the instrument is steady. Only a human-coded set says it is pointed at the truth.

What to Report: One Validation Table

One table per LLM-coded variable, in the appendix, before the reviewer asks for it.

Check	Type	What to report	Hormuz
Existence check	Reliability	Share of quotes unverifiable on the archived page	✓ 2.0% (58 of 2,910)
Prompt stability	Reliability	Statement overlap across 2 to 3 query variants	Target >70%
Inter-model κ	Reliability	Cohen's κ against a second model family	Target $\kappa > 0.70$
Held-out human set	Validity	n , strata, per-class κ or F1	✗ Not yet
Recall audit	Validity	False-silence rate from a hand search of silent cases	✗ 89 to sample

For the qualitative-minded, in Lincoln and Guba (1985) terms: the existence check and archived sources give *confirmability*; prompt stability and a pinned model, *dependability*; a second coder or peer-debriefer agent, *credibility*; a scope statement, *transferability*.

Rows one to three: the pipeline agrees with itself. Rows four and five: it agrees with the world.

Three More Uses

- **Policy existence at scale.** Collect every budget bill from the EU member states → code whether a policy exists → find its correlates.
- **Needle in a haystack.** Structured web searches, in several languages, for the one piece of qualitative evidence you need → then follow up by hand.
- **Existing corpus, new question.** Take ParLEE, filter by domain, run agents to find the relevant speeches → code → correlates → models.

Same skeleton every time: retrieve, archive, extract, verify. Only the corpus and the codebook change.

Surveys That Listen Back

The traditional survey is a script.

The questionnaire is fixed. The respondent's answers cannot reshape the next question.

An agentic survey is a conversation.

- An LLM reads the participant's open-ended response in real time.
- It decides what to ask next from a pool of candidate items.
- It can *create* new items from earlier responses.
- It can probe ambiguity, ask for examples, or pivot.

The survey adapts to the respondent — not the other way around.

Why this is hard with classic methods

- Branching logic is brittle and pre-specified.
- Open-ended responses are coded *after* the fact.
- Salient issues that researchers didn't anticipate go unmeasured.

LLMs solve all three problems at once.

The adaptive-survey programme this section builds on: Velez (2025); Velez & Liu (2025); Velez, Liu & Clifford (2026).

A Ladder of Adaptivity: Who Writes the Next Question?



Velez (2025, *Political Analysis*); Velez & Liu (2025, *APSR*); Velez, Liu & Clifford (2026, *APSR*); Velez & Liu, *Tailored Experiments* (Cambridge Elements, forthcoming).

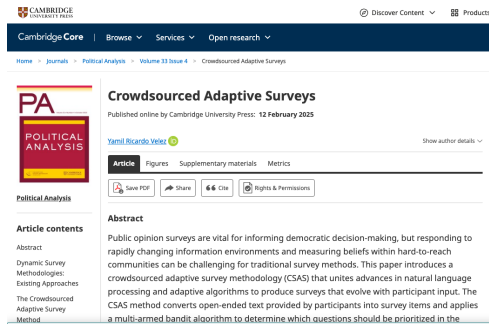
Rung 4 is *agentic* only with a goal, a tool such as retrieval, and a stop rule.
Otherwise it is a chatbot with a long memory.

Rung 1: Crowdsourced Adaptive Surveys

The respondents co-write the questionnaire.

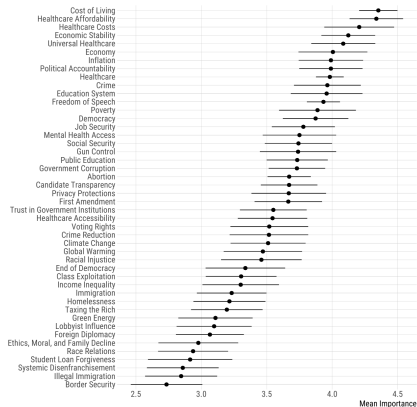
- 1 Ask an open-ended question: “an issue you care deeply about.”
- 2 An LLM rewrites each answer as a closed survey item.
- 3 Filters drop near-duplicates (embedding similarity > 0.90) and toxic items.
- 4 Each respondent rates their own item plus k items from the bank.
- 5 A bandit (Gaussian Thompson sampling) picks what the *next* respondent sees.

LLM cost: a fraction of a cent per respondent. Fits inside a normal survey as a few “dynamic” slots.



Velez (2025), *Political Analysis*, open access.

Rung 1: What It Found



Velez (2025), Figure 2: mean importance, crowd-written items.

Issue importance ($N = 820$)

Seeded with 8 Gallup items; the crowd's items outranked the seeds. Surfaced: mental-health access, privacy, candidate transparency.

The bandit is the catch: the probability of being shown depends on the data, so raw means over-weight early winners. Figure 2 is inverse-probability weighted.

Latino misinformation ($N = 319$)

Seeded with fact-checkers' claims. The most-believed claims came from participants.

Inductive instrument-building at survey scale.
Pre-register the pilot/main split.

Rungs 2–3: Tailored Experiments



Rung 2: the stimulus adapts (Velez & Liu, APSR 2025)

Five experiments, $N \approx 1,100$ –2,000 each.
Counter-arguments aimed at your *own* core issue produced moderation, not backlash. Polarisation appeared only when the argument was vitriolic.

Rung 3: the conversation measures (Velez, Liu & Clifford, APSR 2026)

A GPT-4o dialogue asks *why* you hold the view and extracts the **focal belief**. Arguments aimed at it beat arguments aimed at a plausible-but-unmentioned belief on 11 of 12 outcomes. Only the focal effect survived at 10 days.

The interview transcript is no longer just data. It is the input to the treatment.

The Catch: Every Respondent Gets a Different Treatment

The estimand moves.

The effect is no longer “of this text.” It is the average effect of *a text aimed at your belief*, whatever the generator wrote.

Formally: $T_i = g(\text{elicited text}_i)$ and the estimand is $E[Y(g) - Y(g_{\text{placebo}})]$, the effect of the generator *policy* against a placebo policy, not of any one text. Say so in the paper.

So the audit trail has to carry the treatment.

- Log every generated stimulus. Hash the prompt. Pin the model. When the model is retired, g cannot be re-run; only the archive remains.
- Placebo-control against a text corpus, not against nothing.
- Manipulation check: did the focal argument move the focal belief *more*? It did.
- Quality check against human-written arguments.
- Content filter, disclosure, debrief, pre-registration.

Same discipline as Stage 3 of Hormuz: the model may *generate*, but everything it generates is archived and checkable

The Same Machinery Persuades

Issue-based microtargeting (Velez & Liu, under review)

LLM-written audio ads tailored to a respondent's *own* priority issue raised candidate support by over 10 points against the best generic 2024 ad. That is about a quarter of the effect of shared partisanship.

Tailoring to demographics or Big Five traits: mixed, small.

What this means for the interviewer on Rung 4

An adaptive probe is a tiny tailored treatment. “Leading effects” are not a bug to prompt away. They are the mechanism Rungs 2–3 exploit on purpose.

So measure them: randomise probe wording, log every turn, report sensitivity.

The Field Is Splitting Into Three Camps

- ➊ **AI-led interviews at scale.** The LLM runs the whole semi-structured interview with adaptive follow-ups.
Geiecke & Jaravel, “Conversations at Scale”; Wuttke et al., “AI Conversational Interviewing”; **Anthropic Interviewer** (1,250 → 81,000 interviews, 159 countries, 70 languages).
- ➋ **AI-assisted human interviewing.** Human stays primary; LLM suggests probes, flags missed themes, supports note-taking.
Zhang et al. (real-time follow-up generation); Wen et al. (“unobtrusive co-interviewer”).
- ➌ **Agentic / multi-agent systems.** The interviewer plans, tracks coverage, simulates conversational rollouts, updates strategy mid-interview, sometimes codes downstream.
SparkMe (Anugraha, Padmakumar & Yang) — interview as an optimisation problem; Panfilova et al. (2026) — evaluation across 6 frontier models.

The seductive claim: depth-of-interview at survey scale.

The hard problem: “depth” is rapport, trust, silence, embodiment, reflexivity — not just longer text.

Rung 4 With a Leash: An Agent That Must Cite Its Library

MIDEBT. Drafted for OSF registration; not yet registered.

The puzzle. Publics endorse debt reduction and refuse to pay for it. Asked unprompted what to do, they name a *symbolic policy*: salient, other-paying, morally satisfying, and far too small to close the gap.

Across coded open-ends in Japan, Italy and Brazil ($n = 5,037$), the costly option is named by 3 to 10%.

The design. The respondent writes their own fix *before* randomisation. Then an LLM interviewer takes it up. ... ELICIT →

[RANDOMISE] → CONVERSATION

→ BUFFER → OUTCOMES ...

Arm	What the interviewer does
A <i>placebo</i>	Scripted off-topic chat, one verified figure. Zero fiscal content.
B <i>inform</i>	Two mandatory country facts about the size and persistence of the adjustment. Never evaluates the respondent's proposal.
C <i>challenge</i>	Same two facts, then concedes the direction, prices their own proposal against the annual ask, and asks who covers the rest. Up to five rounds.

$\tau_P = E[Y(C) - Y(B)]$ is the study's quantity: information held constant, pricing and incidence varied.

The conversation is the treatment, so it has to be as auditable as a paragraph of text.

What the Agent Is Allowed to Say

The library is the treatment. Arm C may price only measures for which that country's **signed fact sheet** carries a verified ceiling. Every figure is page-cited to one primary source at one vintage.

Three standing rules

- Always price against the *annual adjustment*, never the debt stock. Against the stock everyone is wrong in the same direction and the measurement dies.
- Where no defensible total exists, say so. **Never invent one.**
- No country opens without a signed fact sheet, a native-speaker pass, and two live conversations read by hand.

Country Measure		Share of the ask
Italy	Chamber of Deputies, €997mn	$\approx 1/22$
Italy	Tax-evasion recovery, €2.2bn	$\approx 1/10$
Japan	45-seat cut, ¥3.4bn	0.018%
Japan	The whole Diet, ¥136.5bn	0.71%
Brazil	Corruption	none exists

The last row is the important one. The modal Brazilian proposal cannot be priced, so the agent says that instead, and the design reports the asymmetry rather than smoothing it.

Did the Agent Follow Its Own Rules?

Arm B was forbidden to comment on the respondent's fix. Arm C had to put a price on it.

A human coder read the interviewer's side of every transcript and scored what it actually did.

What the interviewer did	B	C
Said how big the gap is	100%	100%
Said it lasts for years	100%	100%
Put a number on their fix	24%	56%
Agreed before challenging	0%	97%
Commented on their fix	0%	88%

Arm B never broke its rule: 33 sessions out of 33.

Locked before the first respondent

The engine's code, the three prompts, the model version and its settings, each country's fact sheet, the survey, the coding rules, the debrief text. All hashed, so a later edit cannot hide.

The survey records which arm it assigned. The engine records which arm it ran. The two must match in every session.

The same rule as Hormuz

The model talks freely. It may not introduce a number that is not on the vetted sheet.

Every transcript is kept, so you can check later what any respondent was told.

The agent may search. It may not pretend to know.

What we can defend now

- Large- N *exploratory* interviewing.
- Open-ended survey modules with 1–3 adaptive probes.
- **Asynchronous adaptive interviewing** — the real coordination win: respondent clicks a link, gets tailored probes, no scheduled interviewer.

What we cannot defend

- LLM as *respondent* for primary qualitative data.
(Kapania et al., “Simulacrum of Stories.”)
- LLM as *interpreter* — it applies the codebook, doesn’t write it.

Two risks the field hasn’t reckoned with

Re-identification. Researchers re-identified participants in a public interview dataset using agentic web search. Rich transcripts are quasi-identifiers.

Leading effects. LLMs are trained to be helpful. They over-affirm, normalise assumptions, ask subtly leading probes — worst on sensitive topics.

The honest framing:
adaptive open-ended measurement at survey scale.

Step 1: Simulating Respondents (“Silicon Samples”)

The idea. Prompt an LLM to take on a persona (age, income, party, region) and answer a survey *as that person would*.

Good for

- Pre-testing instrument wording.
- Sketching expected effect sizes.
- Stress-testing designs cheaply.

Not good for

- Estimating actual population means.
- Heterogeneity the training data underrepresents.
- Replacing real respondents in published estimands.



Political Analysis

Article contents

Out of One, Many: Using Language Models to Simulate Human Samples

Published online by Cambridge University Press: 21 February 2023

Lisa P. Argyle , Ethan C. Busby, Nancy Fulda, Joshua R. Gubler , Christopher Rytting and David Wingate

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Synthetic Pennsylvania samples match the marginals, which raking does for free. The information is in the joint distribution and the variance, and silicon samples compress both. Argyle et al. (2023); Bisbee et al. (2024).

Step 2: The Building Block

A persona + a prompt = a synthetic respondent.

A persona + a defined interest = an actor in a game-theory model.

What If the Personas Interact?

The personas in Steps 1 and 2 are *static*.
They answer; they don't *interact*.

What changes when they do?

That is the next step.

Step 3: Generative Agent Societies (Park et al. 2023)

Populate a virtual town with 25 LLM-powered agents — each with a persistent identity, memory, and goals — and let them *interact*.

- Agents **remember**, **reflect**, and **plan**.
- Emergent social behaviour follows: they organise a party, spread information, form relationships — without being told to.
- Architecture: perception → memory stream → retrieval → reflection → action.

This is where *agent-based modelling* meets LLMs.

The Natural Extension: Agentic War Games

Same architecture, raised stakes:

- **Crisis bargaining.** Iran, Israel, US, allies as separate agents with mandates and red lines — run the next 30 days a thousand times.
- **Legislative coalition formation** under counterfactual rule changes.
- **Information ecosystems.** How does a deliberative public respond to a misinformation shock?

Used today in defence and policy war-gaming — with all the validity caveats that follow.

The validity warning

Agents behave like *the median text on the internet*, not necessarily like humans.

Treat outputs as **hypothesis-generating** or as a stress test of your own theory — **not as evidence about the world.**

One Project Through the Lifecycle Lens

The Hormuz pipeline. Write and Submit rows are the plan, not the record.

Stage	The agent does	I do (the gate)
Ideate	Pulls the literature on district exposure and war rhetoric	Choose the puzzle: costs are uneven, so should opposition be?
Design	Drafts codebook v0 from the construct	Hand-code cases; write the “if ambiguous, do X” rules; lock and hash the prompt
Build	Downloads BLS and Census, builds the crosswalk, runs 538 retrieval agents	Write the spec and the stopping rule; read what changed; run the cost audit
Analyse	Codes 2,852 quotes; runs the models	Check the failure register and the 2.0% failure rate; interpret the null
Write	Drafts sections in my voice via a style skill	Own every claim, every number, every citation
Submit	Mock referee report; journal comparison	Decide what to fix and where to send it

I was the bottleneck on purpose. Where I was not, I could point to an artefact that let me check.

Analyse: Robustness and Simulation With Human Permissions

What the agent runs

- The robustness set: alternative specs, samples, measures, placebo tests.
- Simulations: silicon samples for pre-tests, agent societies for stress-testing theory.

Three permissions it does not have

- ① **To touch the headline.** The pre-registered spec is frozen.
- ② **To spend.** A hook in the runtime, not a prompt, blocks any batch job without an approved cost audit.
- ③ **To delete.** Every specification goes to a ledger it can add to but never edit.

What you evaluate: the specification curve

The ledger is a specification curve that cannot be pruned: the garden of forking paths, with the paths logged.

Read the distribution of estimates, not the summary. Ask what was *not* run.

Simulations are hypothesis-generating. The permission structure keeps them from becoming evidence by accident.

Analyse, Qualitative: The Agent Applies the Codebook

Deductive coding at scale. Same three checks as Lesson 2: existence, prompt stability, inter-model κ .

Inductive first pass. The agent proposes candidate themes from a sample; you keep, merge, rename. Newer set-ups add a second agent that *argues with the first* before you see the themes.

Agent-as-Peer-Debriefing (2026); multi-LLM thematic analysis with κ + semantic similarity (2025).

Reflexivity. The prompts and the model are part of the reflexive account. Log them; report them.

Ibrahim & Voyer (2026), *Qualitative Research*:
“technological reflexivity.”

Tools

qualmer (R): AI-assisted coding with κ and F1 built in.

Ollama + an open model: transcripts that cannot leave the building.

NVivo, ATLAS.ti, MAXQDA: AI assistants exist; none is agentic yet.

The line

The agent applies the codebook. It may draft v0; it does not own the rules, and it does not interpret. The interpretive act is the comparative advantage worth keeping.

Write: Your Voice as a Skill

Refining prose is the easy win. Do it with a leash.

A **writing-style skill** is a text file the agent loads before it touches prose: sentence rhythm, hedging rules, how you open a paragraph, how you cite, and a list of the tells of machine prose to strip.

Three rules

- The agent may edit *style*. It may not touch claims, numbers, or citations.
- You see every edit before it lands. Accept or reject sentence by sentence.
- Run the de-AI pass last. It removes the tells; it adds nothing.

The journal's guidelines are a skill too

Word limits, abstract structure, replication policy, AI-disclosure wording. Encode once; run it against every draft.

Disclose

Cambridge (APSR): AI may not be an author; declare its use as you would any tool. With an audit trail you can say *exactly* what it did.

The voice is yours. The typing is not.

Write: The Manuscript as Data

Text-analysis tasks the agent does better than a tired coauthor.

- **Definition drift.** List every place the key construct is defined; flag the differences.
- **Term and notation consistency** across sections written by different coauthors.
- **Structure alignment.** Does each section deliver what the introduction promised?
- **Claim–citation check.** Retrieve the cited paper. Does it say that?
- **Table–text match.** Every number in prose equals the number in the table.

Output: a list of flagged locations with the evidence.
The agent *never* rewrites the flagged passage unasked.

In MStack: `/coauthor-review`.

Why this matters for qualitative work

Interpretive papers live or die on consistent use of concepts. Drift between the theory section and the empirics is the most common referee complaint, and the cheapest to catch.

The agent finds. The coauthors decide.

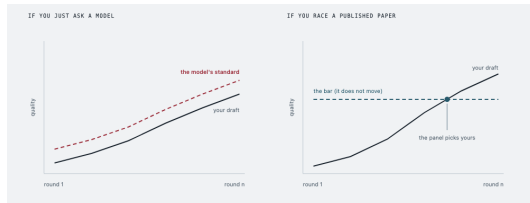
Submit: Run the Journal Before the Journal Runs You

Journal fit and guideline check

Word count, abstract format, replication policy, disclosure wording. Mechanical; let the machine do it. `/journal-fit` compares three journals and logs what it read.

Mock referee

A reviewer-voice skill built from your own referee reports writes the R1 report before submission. It front-runs the objection that would have sunk the paper. (`/referee-mock`)



Ask a model if the draft is good and its standard climbs with you. Race a published paper and the bar stays put.

The mock referee is a *stress test*, not a verdict. Some of what it flags is wrong. The human decides which.

Ship the paper *you* would defend, having heard the objections early.

Manage: Version Control Is the Lab Notebook

Put the agent inside git.

Every action it takes becomes a commit with a message, on a branch. Before it reaches the main line you read the *diff*: the line-by-line list of what changed.

- One branch per analysis; one working copy per parallel agent.
- Checkpoints and resume (Lesson 1) fall out for free.
- Lock the headline (`/freeze`); archive the run (`/archive`); write the retrospective (`/retro`).
- Overleaf users: sync the project to git. The agent edits the `.tex`; you review the diff.
- Paste a draft into a GPT and you get a manuscript back. Put the agent in git and you get the diff.

Even this deck

The September frames were added by an agent: marked with a comment, backed up, compiled, reviewed page by page, and trimmed where they overflowed.

Diff, not summaries.

“I read the diff” is the whole method.

The Tools You Actually Need

In R

- **ellmer** — Call any LLM API (OpenAI, Anthropic, Google, Ollama). Structured output, tool calling, batching. Tidyverse-native.
- **quallmer** — AI-assisted qualitative coding with κ and F1 baked in.

In Python

- Official SDKs from OpenAI, Anthropic, Google.
- Structured-output helpers (Instructor, Pydantic).
- Batch APIs from every major provider (50% discount).

Agent runtimes you can use today

- **Claude Code, Codex, Gemini CLI** — a research task, come back later. The first two run inside their own desktop app.
- **Cursor, Windsurf, Positron** — IDEs that read your whole project folder.
- **Custom GPTs, Claude Projects** — standing context in a chat window. No hooks or logs.

Local (sensitive data)

- **Ollama** + open weights. 16 GB RAM opens most options.

These days I rarely leave two windows: Claude and ChatGPT.

Do You Need to Train Anything?

Five ways to make a model “know your field,” cheapest first.

- ① **Prompt.** One-off instructions. Fine for a pilot.
- ② **Project memory.** A standing file the agent reads every session: data, codebook, conventions.
- ③ **Skill.** A reusable role with rules and worked examples: your writing voice, your referee voice, a coding protocol.
- ④ **Retrieval over a corpus you curate.** Your papers, the subfield’s key papers, the journal’s guidelines, a vetted fact library. Read at run time; nothing is trained. *This is what a custom GPT does.*
- ⑤ **Fine-tuning.** Rare. Narrow classifiers only: Velez (2025) fine-tuned a small model to filter falsifiable claims in CSAS.

A custom GPT: a standing prompt over a curated corpus. Nothing trained.

The small corpus keeps the citations inside the field. Put that corpus in a harness and you get three things the chat window does not: a log of what it retrieved, a diff for every edit, and a hook that can stop it. Sensitive corpus? Ollama on a laptop.

Anatomy of a skill

Name; when to trigger; the role and voice; hard rules; worked examples; output format. A text file. It travels with you.

MStack is a library of ~34 of these for the lifecycle.

Lesson 3: Do a Cost Audit

Where the Hormuz study actually spent money:

Task	Calls	Cost
Stage 1: pairwise BT ranking	45,451	\$4.50
Stage 3: retrieval (LLM + tool)	1,498	\$49.91
Stage 3: extraction (no internet)	2,549	\$5.60
Stage 4: coding	2,852	\$2.57
Total	52,350	\$62.58

Comparable RA-coded study: $\$15/\text{hr} \times 100 \text{ hr} \approx \$1,500$.
Pipeline is $\sim 24\times$ cheaper.

What actually happened

Audit projection: **\$52.07**. Actual: **\$58.07** on rhetoric stages — **11.5% over**.

The dry-run on 3 hand-picked legislators systematically *underestimated* escalation depth.

Lessons

- Pilot on a *stratified* sample, not on easy cases.
- Wire a real-time **kill-switch**, not just a pre-run gate.
- Make resume-from-checkpoint first-class.

Tips and Tricks From the Trenches

Run it on the subscription first

Subscription tokens are heavily subsidised next to API rates, and Claude Code and Codex both run on them.

Add an **API key** only for what a subscription cannot do: a batch submission, or calls from your own script. The harness is then the *outer* agent running an *inner* fleet.

Have Codex and Claude check each other

Different model families = free **inter-model reliability**.

One drafts, the other reviews. One codes statements, the other recodes. Report κ .

Use the Batch API

Every major provider offers it: **50% discount**, results within 24 hours.

Indispensable above a few thousand calls.

Pilot first (~ 100 calls); read the usage field; project; *then* submit.

Cost guardrails (non-negotiable)

Agents will happily spend \$500 in 4 minutes.

Turn off auto-recharge. Hard stop when credits run out.

Add a **hook** that blocks batches without an approved cost audit.

The Cost Hurdle: The Harness Is Free, the Model Can Be

1. **The harness.** Open source and free: **pi**, OpenCode, Codex CLI, Goose, Aider, OpenHands. You pay only for tokens.
2. **The model:** four ways to pay.
 - **A subscription.** A monthly plan carries the coding agent and its tokens, heavily subsidised next to API rates. Start here.
 - **Frontier, closed, by API.** Sonnet 5: \$2 in / \$10 out per million tokens, roughly 750,000 words. Opus 5: \$5 / \$25.
 - **Open-weight, by API.** GLM-5.3: \$1.4 / \$4.4. GLM-5.3-Flash: \$0.075 / \$0.25 (promo). Same harness, one config line.
 - **Open-weight, local.** Ollama or LM Studio on your machine. Zero marginal cost, data never leaves, weights can be archived.

Prices as of 7 Sept 2026. Batch API halves the closed-model figures.

Plan with the expensive model. Run the loop with a cheaper one.
Then check that the swap did not change your codes.

pi: A Harness You Can Read in an Afternoon

Mario Zechner's pi (MIT licence)

- Four tools: read, write, edit, bash.
- Prompt plus tool definitions under 1,000 tokens.
- Any provider or OpenAI-compatible endpoint, so local models work. Switch models mid-session.
- No plug-in protocol, no sub-agents, no plan mode. On purpose.

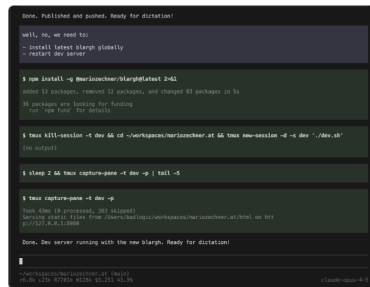
Two cautions

Permissions. pi runs unrestricted by default. Add permission checks before the API key.

Where the money went. In Hormuz, 80% of the bill was retrieval (LLM plus search tool). Open weights cut the model line only.

What I learned building an opinionated and minimal coding agent

2025-11-30



```
Done. Published and pushed. Ready for dictation?

well, no, we need to:
- install latest blargh globally
- restart dev server

$ rpm install -g @mariozechner/blargh/latest 2-61
added 13 packages, removed 12 packages, and changed 83 packages in 5s
36 packages are looking for funding
  run 'rpm fund' for details

$ tox kill-session -t dev && cd ~/workspaces/mariozechner.at && tox new-session -d -s dev './dev.sh'
(no output)

$ sleep 2 && tox capture-pane -t dev -p | tail -5

$ tox capture-pane -t dev -p
Took 43ms (8 processed, 353 skipped)
Serving static files from /Users/badlogic/workspaces/mariozechner.at/html on http://127.0.0.1:3000

Done. Dev server running with the new blargh. Ready for dictation!

~/workspaces/mariozechner.at (main)
r6.0a 123b 87781b W123b 65.251 43.3s  c laude-gpt4o-4-5
```

It's not much, but it's mine

mariozechner.at, 30 Nov 2025. "It's not much, but it's mine."

The Mindset Shift

You are no longer the typist. You are the manager.

- Give the agent context: data, codebook, draft, constraints.
- Specify the deliverable as if briefing an RA.
- Demand checkpoints and a log.
- Read the diff. Question the choices.
- Validate, validate, validate.

The agent is patient, fast, cheap, consistent, and *wrong in interesting ways*.

Your job is to design the workflow that catches the wrongness before it reaches the table in the paper.

What hasn't changed

- Construct validity is still on you.
- Identification is still on you.
- The theory is still on you.

What has

The cost of running an ambitious empirical project alone has fallen by an order of magnitude.

The set of people who can do that work has expanded.

Economics

Coding agents in the social sciences

May 27, 2026

Anthropic, May 2026: survey of 1,260 social scientists

81% have tried chatbots. 20% use coding agents. Twice as many researchers with typically male names use them; top-university researchers are 40% more likely to.

The dividend goes to early adopters. The gaps are already forming.

Rules of the road

- AI cannot be an author (Cambridge, *APSR*); your university and examiners have policies too. Declare use as you would any tool.
- A chat window hands back a manuscript. A harness hands back the diff and the log.
- Your thesis outlives the model. Archive open weights or budget a re-validation.

What transfers

Writing a codebook. Git and reading a diff. Designing a validation sample. Fluent output is not judgment; these are.

The set of people who can do ambitious work alone just got larger. Be in it.

Shameless Plug: Scaling Open-Ended Survey Responses

JOURNAL ARTICLE

Scaling Open-Ended Survey Responses Using LLM-Paired Comparisons

Matthew R DiGiuseppe ✉, Michael E Flynn

Public Opinion Quarterly, nfag013, <https://doi.org/10.1093/poq/nfag013>

Published: 27 March 2026

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Abstract

Survey researchers rely heavily on closed-ended questions to measure latent respondent characteristics like knowledge, policy positions, emotions, ideology, and various other traits. Closed-ended questions are easy to analyze and collect, but necessarily limit the depth and variability of responses. Open-ended responses allow for greater depth and variability in responses, but are labor intensive to code. Large language models (LLMs) may help with this problem, but existing approaches to using LLMs have a number of limitations. In this paper, we

DiGiuseppe & Flynn (2026).
Public Opinion Quarterly.

The problem: open-ended survey responses are rich, but coding them by hand doesn't scale.

The method: **LLM-paired comparisons**.
Ask the model “which respondent shows more economic knowledge?” thousands of times; recover a latent score via Bradley–Terry.

Same agentic primitive — pairwise LLM scaling — that Stage 1 of the Hormuz pipeline uses to rank industries.

doi.org/10.1093/poq/nfag013

Shameless Plug: MStack

A Claude Code plugin for academic research.

~34 slash commands that walk a paper across every stage — with the *right forcing questions* at the right moment.

Stage	Example commands
Ideate	/research-question, /scope-challenge
Map	/lit-map, /theory-build
Design	/identification-review, /preregister
Build	/data-acquire, /codebook
Analyse	/robustness, /results-audit
Write	/draft-section, /abstract-shotgun
Submit	/journal-fit, /referee-mock
R&R	/r-and-r, /coauthor-review

Design principle

Role > prompt.

Every skill speaks in a defined voice — advisor, methodologist, referee, replicator. The forcing questions are baked in.

Install (two lines in Claude Code)

```
/plugin marketplace add  
  matthewdigiuseppe/MStack  
/plugin install mstack@mstack
```

github.com/matthewdigiuseppe/MStack

Built for IPE. Generalises to most quantitative social science.

Ethics, Part One: The Worry That Is Overstated

Keep the footprint in proportion

- One text prompt is small: seconds of a microwave, drops of water. Agentic work is heavier, but stays in the range of ordinary domestic media use.
- Per-prompt energy is falling fast as models and hardware improve.
- Data centres are a small, fast-growing share of world electricity. That growth is a grid and siting question, not a question about your prompt.

If it still bothers you, offset it.

Watch an hour less video. Take the train. Turn the thermostat down. Those levers move real numbers.

Sources: Google (2025); IEA (2025).

Why abstaining is the wrong lever

- Your use is a drop in the ocean. The train is not stopping for anyone.
- If everyone else gets more productive and you do not, you have made a point nobody will hear, and paid for it yourself.
- The employer's calculation: familiarity with these tools is becoming a hiring assumption.
- The right lever is policy: grid, siting, cooling, disclosure. Push there. Work here.

Abstaining is a cost you pay, not a cost you prevent.

Ethics, Part Two: The Worries That Are Not Overstated

- **The people in your data.** Rich transcripts re-identify. Transcripts sent to a US API need the DPIA and the ethics clause first.
- **Persuasion at scale.** The tailoring that measures better also persuades better. An adaptive probe is a treatment.
- **Accountability.** An AI cannot be an author because it cannot be accountable. Declare use; keep the audit trail that lets you declare precisely.
- **Hidden labour.** The RA who quietly runs everything through ChatGPT, and the annotators behind the models.
- **Uninvited agents.** Bots now fill in surveys and turn up to online interviews. Design for it.

The ethics question is not whether to use the tool.

It is what you owe the people in your data, and whether you can show your work.

Agents can draft. They cannot answer for the claims. You can.
They will let you write a paper you couldn't write alone.

The methodological work — audit trails, validation, honest construct definition —
is what makes the difference between a press release and a publication.

Thank you.

m.r.digiuseppe@fsw.leidenuniv.nl

Questions?

Your First Pipeline in a Weekend

Six steps, and the last one is a methods appendix.

- 1 Take 200 texts you already have.
- 2 Write codebook v0 on 50 hand-coded cases. Hold out a separate random set you never touch while prompting.
- 3 **ellmer** with structured output: one model, one prompt.
- 4 κ against your codes, reported per category.
- 5 Recode with a second model family. Report both.
- 6 Write it up as a validation note. That is a methods appendix.

Before any transcript leaves the building: the DPIA, the data-sharing clause in your ethics approval, and the provider's retention terms.

The access ladder

- 1 A university licence. Ask; there may be one.
- 2 A monthly subscription that includes a coding agent.
- 3 An API key with a hard cap and auto-recharge off.
- 4 Ollama and an open model when data cannot leave.

Cheap to learn now. Supervisors will start expecting it.

Inside the Retrieve Call: The Tier 0 Prompt

T0 is one of **five** structurally identical retrieval prompts (T0–T4). If T0 yields 0 verified quotes, the pipeline fires a *fresh agent call* with the T1 prompt; same for T2–T4.

You are a research assistant locating public statements made by US legislators about the February 28, 2026 US military operations against Iran and the Strait of Hormuz crisis.

Your job in this call is RETRIEVAL ONLY. Issue web searches via the `web_search` tool, then return a JSON object listing the queries you issued and the URLs you surfaced. You **MUST NOT** extract quotes, summarize articles, paraphrase content, or write any prose beyond the JSON output.

TASK -- TIER 0 (CANONICAL)

Issue exactly TWO search queries:

1. "{full_name}" Iran Hormuz 2026
2. "{full_name}" Iran statement 2026

Both queries **MUST** contain "2026" to filter out 1979 / 2015 coverage.

OUTPUT -- STRICT JSON, NO PREAMBLE:

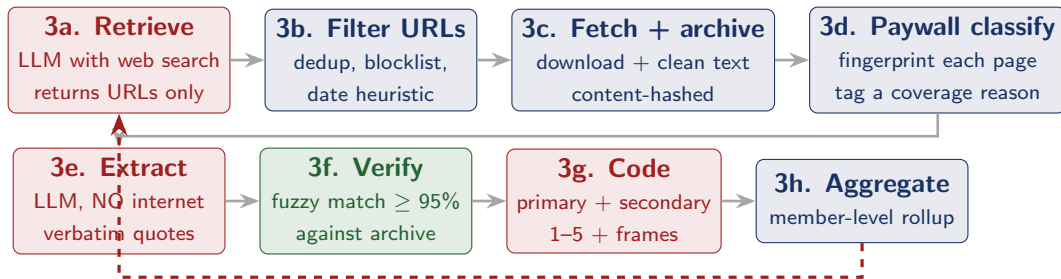
```
{
  "queries_issued": ["string", ...],
  "urls_found": [{"url": "...", "title": "...",
    "snippet": "...verbatim from search tool...",
    "rank": 1, "appeared_for_query": "..."}, ...]
}
```

CONSTRAINTS

- Do NOT include URLs from before 2026-02-21.
- Do NOT extract or quote article content.
- Snippets must be verbatim from the search tool, not your summary.

The Rhetoric Pipeline in Detail

SWARM: 538 agent runs — one per legislator, 8 running in parallel



LOOP (run by the pipeline): if 0 verified quotes $\rightarrow$ escalate tier ($T_0 \rightarrow T_1 \rightarrow \dots \rightarrow T_5$ stop)

■ LLM judgment ■ deterministic code ■ mechanical verification — loop

Concurrent Work Worth Citing

Methodologically related papers that landed alongside this pipeline:

Afonso, Galiani, Gálvez & Sosa (2026)

Deep Research on a Loop: Using AI Agents to Construct Economic Datasets.

NBER Working Paper

Proposes **DRIL** — two-stage agent architecture (design → implementation), frozen protocol, citation contract, explicit data-gap taxonomy. Applied to the Global Tax Expenditures Database for eight LAC countries. Strong overlap with Stage 3 of this pipeline; we add mechanical quote verification and a strict retrieval/extraction split.

Poast (work in progress, University of Chicago)

LLMs and Data Management for International Relations Scholars:

Moving Beyond EUGene-Style Software. [SSRN](#)

Positions LLM-based pipelines as the successor to EUGene-era IR data-management tools — same architectural family, different substantive domain. The convergence is the point.

A Reading List

Adaptive surveys and tailored experiments

- Velez (2025). Crowdsourced Adaptive Surveys. *Political Analysis*.
- Velez & Liu (2025). Confronting Core Issues. *APSR*.
- Velez, Liu & Clifford (2026). Beyond Belief Change. *APSR*.

Validation and inference with LLM labels

- Pangakis, Wolken & Fasching (2023). Automated Annotation with Generative AI Requires Validation.
- Gilardi, Alizadeh & Kubli (2023). ChatGPT outperforms crowd workers. *PNAS*.

Silicon samples and agent societies

- Argyle et al. (2023). Out of One, Many. *Political Analysis*.
- Bisbee et al. (2024). Synthetic Replacements for Human Survey Data? *Political Analysis*.
- Park et al. (2023). Generative Agents. *UIST*.

Agents building datasets

- Afonso, Galiani, Gálvez & Sosa (2026). Deep Research on a Loop. NBER WP 35188.
- Kohler, Zollikofer, Einsiedler, Hoyle & Ash (2026). Read the Paper, Write the Code. *arXiv*.

Adoption

- Anthropic (2026). Coding agents in the social sciences.

Building the District Exposure Score

Step 1: Industry vulnerability score.

Pairwise LLM comparisons across 302 NAICS industries:

"Which suffers greater net harm from a Hormuz closure?"

Bradley–Terry recovers $\hat{\beta}_i$.

Step 2: Employment shares.

BLS QCEW 2023 county $\times$ 4-digit NAICS counts E_{ci} .

Step 3: County $\rightarrow$ district crosswalk.

TIGER 2025 shapefiles; area-weighted allocation to congressional district d , giving E_{di} .

Step 4: Weighted exposure index.

$$\text{Exposure}_d = \sum_i \frac{E_{di}}{E_d} \cdot \hat{\beta}_i$$

Most exposed districts:

District	Dominant industry	z
TN-04	Resin/synthetics mfg	+2.59
OH-05	Resin/synthetics mfg	+2.33
CO-08	Resin/synthetics mfg	+2.05
TX-14	Resin/synthetics mfg	+1.99

Least exposed districts:

District	Dominant industry	z
NY-12	Financial services	−3.75
DC-98	Financial services	−3.74
NY-10	Financial services	−3.57
CA-11	Financial services	−2.98

The agent downloaded BLS & Census files, parsed NAICS, built the crosswalk, computed the index, wrote the script. *I read the diff.*

Inside the Coding Call

You are coding a single quote attributed to a US legislator about the February 28, 2026 US military operations against Iran and the Strait of Hormuz crisis. You produce structured codes only.

You DO NOT have internet access. Use only the information below.

QUOTE

```
Verbatim text:      {verified_text}
Surrounding context (<=200 char): {context}
```

TASK -- code on the following dimensions:

1. conflict_support -- integer 1-5
 - 1 = strongly opposes / calls for withdrawal
 - 2 = opposes / urges restraint / questions rationale
 - 3 = neutral / procedural / declines to take a position
 - 4 = supports / endorses the operation
 - 5 = strongly supports / calls for escalation
2. economic_cost_frame -- bool
3. security_frame -- bool
4. mentions_constituent_impact -- bool
5. criticizes_administration -- bool
6. praises_administration -- bool
7. calls_for_diplomacy -- bool

ABSOLUTE RULES

- Code only what the quote actually says.
- Do NOT infer the legislator's general position from prior knowledge of them.
- If genuinely ambiguous, set conflict_support to 3.
- Output JSON only.